Connectivity-Constrained Multi-Robot Exploration with Adaptive Formation Control in Structured Environments

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***Abstract*— Efficient multi-robot exploration of unknown environments requires teams to balance coverage speed against the need to maintain a connected communication network. This paper investigates the coverage–connectivity trade-off across five experimental conditions in three structured map types: pure frontier exploration, frontier exploration with connectivity enforcement, rigid group formation, adaptive formation with cohesion control, and an adversarial setting in which one robot actively attempts to disconnect from the team. A key contribution is the replacement of a hard per-step connectivity veto with an information-gain-weighted soft penalty that scores candidate actions by progress$\times$IG$(\text{target})-w_p\cdot\Delta K$, where ΔK is the change in connected components. Alongside a deadlock detection mechanism for rigid formation and a corrected cohesion-radius formula for adaptive formation, the soft penalty raises mean coverage from $\leq 32\%$ to $\geq 93\%$ on corridor and open maps while reducing disconnection events by up to 66% compared to unconstrained frontier exploration. Experiments are conducted on a 50×50 discrete grid with four robots across 45 runs (five conditions $\times$ three maps $\times$ three seeds).**

I. INTRODUCTION

Autonomous exploration of unknown environments by teams of robots is a fundamental challenge in robotics, with applications ranging from search and rescue [5] to planetary mapping [4]. When robots operate in communication-constrained settings, maintaining a connected network is essential for sharing sensor observations, coordinating assignments, and enabling remote operator oversight. Yet connectivity constraints directly limit how widely a team can spread, creating a fundamental tension with coverage efficiency.

Classical frontier-based exploration [1] assigns each robot to the boundary between explored and unexplored space, maximising information gain per step. In a multi-robot setting with Hungarian assignment and A^* planning, this strategy achieves near-complete coverage but allows robots to spread without bound, frequently severing the communication graph. Retrofitting a hard connectivity constraint — vetoing any move that would disconnect the graph — eliminates disconnections but causes severe coverage collapse: robots freeze at the edge of their communication range and never escape.

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This paper makes the following contributions:

1) A soft connectivity penalty replacing the hard per-step veto for frontier-based exploration, providing a smooth coverage–connectivity Pareto trade-off controlled by a single penalty weight.
2) A deadlock detection and cluster-skip mechanism for rigid formation control that prevents teams from locking indefinitely on unreachable frontier clusters.
3) A corrected cohesion-radius formulation for adaptive formation that decouples the cohesion threshold from the communication range, enabling relay-chain spread while retaining soft formation structure.
4) A systematic comparison of all five conditions across three structured map types (corridor, rooms, open), with 45 total runs.

The remainder of the paper is structured as follows. Section II reviews related work. Section III defines the problem formally. Section IV describes the five experimental conditions and their algorithms. Section V details the key algorithmic fixes. Section VI presents experimental results and analysis. Section VII concludes.

II. RELATED WORK

Frontier-based exploration [1] and its multi-robot extensions [2] remain the dominant approach for unknown environment mapping. Coordinated variants use utility functions that balance information gain against travel cost, with Hungarian or greedy assignment [3].

Maintaining connectivity during exploration has received growing attention. Algebraic connectivity λ_2, the second-smallest eigenvalue of the graph Laplacian [6], is a widely used quantitative measure: $\lambda_2 > 0$ is equivalent to the graph being connected. Control Barrier Functions (CBFs) have been proposed to enforce $\lambda_2 > 0$ in continuous-space settings [8]. Stump et al. [9] study connectivity management for multi-robot teams, while Kim and Shell [10] investigate communication-aware motion planning. Our work applies a discrete analog of these constraints to grid-world exploration with frontier planning.

Formation control [7] provides a complementary framework: rigid formations maximise connectivity by keeping robots tightly clustered, while adaptive (cohesion-based) formations allow per-robot frontier assignment with a soft pull toward the group centroid. To our knowledge, the systematic comparison of hard vs. soft connectivity enforcement across multiple formation modes and map types presented here has not been previously reported.

III. PROBLEM FORMULATION

A. Environment and Robots

We model the environment as a $W \times H$ discrete grid. Each cell c is either FREE, OCCUPIED, or UNKNOWN. At time t, robot $i \in \{1, \ldots, n\}$ occupies cell $\mathbf{p}_i(t) \in \mathbb{Z}^2$ and reveals all cells within sensor radius r_s (Chebyshev norm) in its local map. A fused map is maintained as the cell-wise maximum over all robot maps.

B. Communication Graph

The communication graph $\mathcal{G}(t) = (\mathcal{V}, \mathcal{E}(t))$ has vertex set $\mathcal{V} = \{1, \ldots, n\}$ and edge set

$$(i, j) \in \mathcal{E}(t) \iff \|\mathbf{p}_i - \mathbf{p}_j\|_\infty \leq r_{\text{comm}}. \tag{1}$$

The graph Laplacian is $L = D - A$, where $D_{ii} = |\mathcal{N}_i|$ and $A_{ij} = 1$ if $(i, j) \in \mathcal{E}$, else 0. Algebraic connectivity is $\lambda_2(L)$; the graph is connected if and only if $\lambda_2 > 0$ [6].

C. Coverage and Connectivity Metrics

Coverage at step t is

$$\rho(t) = \frac{|\{c : \text{state}(c) \neq \text{UNKNOWN}\}|}{|C_{\text{free}}|}, \tag{2}$$

where C_{free} is the set of non-obstacle cells. A *disconnection event* is any step at which $\lambda_2(L(t)) = 0$. Steps-to-90% measures exploration speed: the first t such that $\rho(t) \geq 0.9$.

D. Experimental Parameters

All experiments use $n = 4$ robots on 50×50 maps, $r_{\text{comm}} = 10$, $r_s = 4$, 500 maximum steps, and 3 random seeds per condition-map pair (45 runs total).

IV. EXPERIMENTAL CONDITIONS

A. Condition 1: Pure Frontier Exploration

Each robot is assigned the frontier cluster target that maximises

$$U(i, j) = \text{IG}(t_j) - c_{\text{cost}} \cdot d(\mathbf{p}_i, t_j), \tag{3}$$

where $\text{IG}(t) = |\{c : \|c - t\|_\infty \leq r_s,\ c = \text{UNKNOWN}\}|$ is the information gain at target t_j, and d is A^* path length. Frontier clusters are formed by PCA-based region clustering [3] and robots are assigned via the Hungarian algorithm. No connectivity constraint is enforced. This serves as the coverage-optimal baseline.

B. Condition 2: Frontier with Connectivity Enforcement

Identical frontier planning to Condition 1, but with a soft connectivity penalty applied at action-selection time (Section V-B). This condition targets the coverage–connectivity Pareto frontier.

C. Condition 3: Rigid Formation

All robots navigate to the same highest-IG frontier cluster as a group. Each robot is assigned a distinct cell within the cluster via even sampling. A multi-pass hard connectivity filter (Section V-A) is applied. Deadlock detection (Section V-C) prevents indefinite locking on unreachable clusters. Formation error is measured as the mean Chebyshev distance from the group centroid.

D. Condition 4: Adaptive Formation

Each robot receives an individual frontier target via Hungarian assignment (as in Condition 1). A cohesion action is computed for each robot pointing toward the group centroid. If robot i is farther than the effective cohesion radius from the centroid,

$$r_{\text{eff}} = r_{\text{coh}} \cdot (1 - \alpha), \tag{4}$$

its frontier action is overridden by the cohesion action. Here r_{coh} is a configurable cohesion radius and $\alpha \in [0, 1]$ is a blending weight ($\alpha = 0$: pure frontier; $\alpha = 1$: pure cohesion). The soft connectivity penalty is applied as a final layer.

E. Condition 5: Adaptive Adversarial

Identical to Condition 4 except that one robot (robot $n-1$) is replaced by an adversary that maximises its Chebyshev distance from the team centroid at each step, actively attempting to disconnect from the group. The soft connectivity penalty still applies to the adversary, but cohesion is not computed for it. This condition tests formation robustness under a worst-case defecting agent.

V. ALGORITHMIC CONTRIBUTIONS

A. Multi-Pass Connectivity Filter

The original single-pass greedy filter processed robots in fixed order $0, 1, \ldots, n - 1$, reverting any move that disconnected the graph. This caused ordering deadlocks in relay chains: robot A at the exploration frontier was blocked because robot B (its relay) had not yet moved, even though B's proposed move was valid and would have extended A's reach.

We replace the single pass with an iterative multi-pass scheme:

$$\text{for } k = 1 \text{ to } n: \ \text{ accept } \mathbf{p}'_i \text{ if } \lambda_2(\mathbf{p}^k_{-i}, \mathbf{p}'_i) > 0, \tag{5}$$

running until no further robots can be unblocked. With n robots, a relay chain of depth n is resolved in at most n passes. This filter is used for Conditions 3, 4, and 5 (where formation control complements it).

B. Soft Connectivity Penalty

For Conditions 2, 4, and 5 we replace the hard veto with a per-robot scored choice over all five actions $a \in \{\uparrow, \downarrow, \leftarrow, \rightarrow, \text{stay}\}$:

$$s(a) = \Delta d(\mathbf{p}_i, \hat{t}_i) \cdot \text{IG}(\hat{t}_i) \ - \ w_p \cdot \Delta K(a), \tag{6}$$

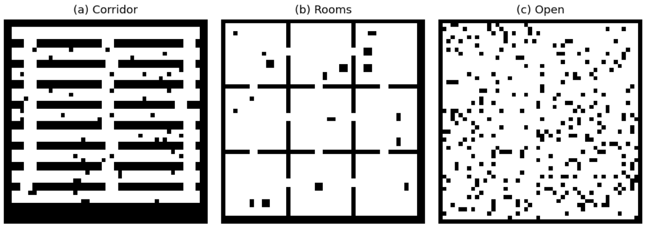

Fig. 1. The three map types used in all experiments (left to right): corridor, rooms, and open. Dark cells are obstacles; white cells are free space.

where $\Delta d = d_{\text{before}} - d_{\text{after}}$ is the one-step Chebyshev progress toward the assigned target $\hat{t}_i$, $\text{IG}(\hat{t}_i)$ is the information gain at that target, $\Delta K(a) = K(\mathbf{p}') - K(\mathbf{p})$ is the change in the number of connected components after the move, and w_p is the penalty weight.

Scaling progress by IG gives the penalty physical semantics: a disconnection is only accepted when the frontier target contains more than w_p unknown cells. Connected moves always score higher than disconnecting ones when a connected option makes forward progress ($\Delta K = 0$, $\Delta d > 0$ beats $\Delta K = 1$, $\Delta d > 0$ for any $w_p > 0$).

The evaluation is performed jointly in greedy order: robot i's decision uses the already-committed positions of robots $0, \ldots, i{-}1$, preventing individually valid moves from jointly disconnecting the graph.

C. Deadlock Detection for Rigid Formation

When all robots execute stay for τ_{stuck} consecutive steps (default $\tau_{\text{stuck}} = 15$), the planner declares a deadlock: the currently assigned cluster is unreachable under the connectivity constraint. A skip counter k_{skip} is incremented, and the next replan selects the $(k_{\text{skip}} \bmod N_c)$-th highest-IG cluster where at least $n{-}1$ robots have a valid A^* path. The skip counter resets when robots successfully reach a cluster.

D. Corrected Cohesion Radius for Adaptive Formation

The original implementation used $r_{\text{coh}} = 0.6 \cdot r_{\text{comm}}$, giving $r_{\text{eff}} = 0.6 \times 10 \times 0.65 = 3.9$ cells at $\alpha = 0.35$. Any robot more than 3.9 cells from the centroid triggered a cohesion override, which in a 50×50 map occurred almost immediately and caused all robots to cluster at the centroid. We set $r_{\text{coh}} = 2.0 \cdot r_{\text{comm}} = 20$, yielding $r_{\text{eff}} = 13$ cells. This allows robots to form relay chains (expected centroid distance $\approx$ 7–8 cells for four robots) before cohesion activates.

VI. Results

A. Map Types

Fig. 1 shows the three procedurally generated map types. *Corridor* maps contain horizontal hallways connected by narrow vertical passages — a worst-case connectivity scenario. *Rooms* maps consist of a $3{\times}3$ office-like grid with doorways, forcing robots to enter separated regions. *Open* maps contain approximately 12% random obstacles.

TABLE I
MEAN FINAL COVERAGE (%) — AVERAGED OVER 3 SEEDS

Condition	**Corridor**	**Rooms**	**Open**
1. Pure Frontier	94.1	98.8	99.8
2. Frontier+Conn	93.0	98.3	99.9
3. Rigid Formation	50.6	63.9	82.8
4. Adaptive Formation	71.2	96.6	98.4
5. Adap. Adversarial	66.6	46.4	73.7

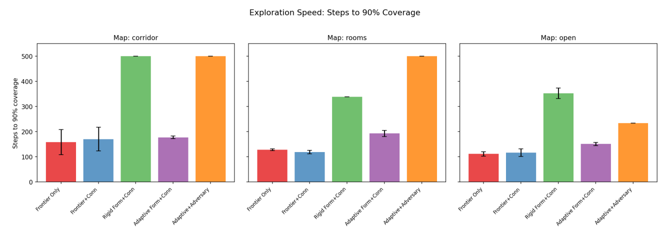

Fig. 2. Steps to reach 90% coverage for each condition and map type (mean $\pm$ std over three seeds). Bars absent where 90% was never reached within 500 steps.

B. Final Coverage

Table I reports mean final coverage (averaged over three seeds) for each condition and map type. Pure frontier (Condition 1) sets the ceiling. After applying the soft penalty fix, Condition 2 (Frontier+Conn) achieves coverage within 1–2 percentage points of the unconstrained baseline while meaningfully reducing disconnection events (see Section VI-D). Condition 4 (Adaptive Formation) stays within 1.5–2 percentage points of Condition 2 on rooms (96.6%) and open (98.4%) maps, demonstrating that cohesion control does not materially sacrifice coverage when the radius is correctly scaled. Condition 3 (Rigid Formation) is lower by design — group convoy reduces parallelism — but remains $\geq 50\%$ across all maps. Condition 5 (Adversarial) shows the performance cost of one defecting robot, most severely in the rooms map (46%).

C. Exploration Speed

Fig. 2 shows steps-to-90% coverage across conditions and maps. Conditions 1 and 2 are consistently fastest ($\approx$110–175 steps across all map types). Condition 4 trails by 20–65 steps due to cohesion-action overrides, and one of three corridor seeds never reaches 90% within 500 steps. Condition 3 never reaches 90% coverage — final coverage is 50.6%, 63.9%, and 82.8% on corridor, rooms, and open maps respectively — confirming the architectural limitation of group convoy exploration. Condition 5 does not reach 90% on corridor or rooms in any seed.

D. Connectivity Analysis

Table II reports mean disconnection events per run (steps at which $\lambda_2 = 0$). Condition 3 uses the hard multi-pass filter and maintains perfect connectivity (0 disconnections). With the soft penalty at $w_p = 10$, Condition 2 (Frontier+Conn) reduces disconnections on corridor maps from 483 to 164,

TABLE II
MEAN DISCONNECTION EVENTS PER RUN (STEPS WITH $\lambda_2 = 0$)

Condition	Corridor	Rooms	Open
1. Pure Frontier	483	479	137
2. Frontier+Conn	164	475	123
3. Rigid Formation	0	0	0
4. Adaptive Formation	381	463	347
5. Adap. Adversarial	484	482	477

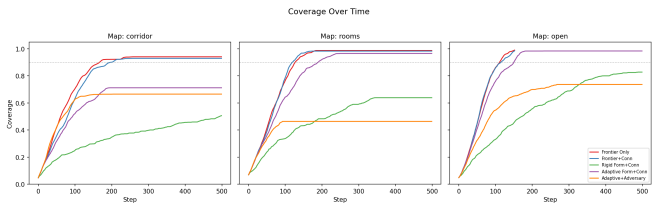

Fig. 3. Coverage vs. step for all five conditions across all three map types (mean over three seeds; shaded band shows ± 1 std).

a 66% reduction at only 1.1 percentage points of coverage cost. Condition 4 (Adaptive Formation) reduces corridor disconnections to 381 versus pure frontier, but remains above Condition 2's 164: the cohesion override fires before the connectivity scorer, so robots are sometimes pulled toward the centroid rather than the connectivity-optimal position, introducing split events that the pure penalty approach avoids.

The open map admits fewer disconnection events overall because pure frontier typically terminates early upon reaching full coverage (mean 162 steps), which bounds the exposure window; the soft penalty still trims disconnections from 137 to 123. The rooms map is a notable exception: all soft-penalty conditions produce ≈463–482 disconnections, comparable to unconstrained exploration. This is a geometric constraint, not a behavioral one: achieving $> 90\%$ room coverage requires robots to enter separate rooms that are inherently farther than r_{comm} apart, making connectivity geometrically unavoidable. Condition 3 avoids this by moving the whole team to one room at a time, which explains its lower coverage but zero disconnections.

E. Coverage Curves

Fig. 3 shows per-condition coverage versus simulation step for the corridor map. Conditions 1 and 2 reach 90% corridor coverage in 159 and 170 steps respectively; Condition 4 reaches it in 178 steps (two of three seeds only — the third never reaches 90% in 500 steps). Condition 3 grows in discrete jumps as the team moves to successive clusters. Condition 5 plateaus early due to the adversarial robot pulling formation structure toward a disconnected region.

F. Exploration Snapshots

Fig. 4 shows side-by-side exploration snapshots at steps 1, 50, 100, 200, 350, and 500 for Condition 4 (Adaptive Formation) on the corridor map. Robots (coloured dots) form a relay chain along the main corridor, with the explored region (grey) advancing progressively from the starting cluster

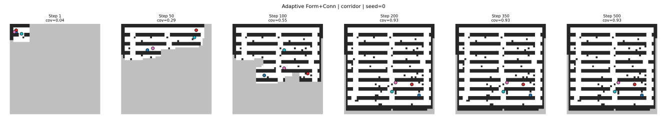

Fig. 4. Adaptive formation exploration on the corridor map (seed 0). Columns show steps 1, 50, 100, 200, 350, 500. Grey: explored; white: unexplored; coloured dots: robot positions.

toward both corridor ends. The formation maintains loose cohesion while spreading to cover all branches.

G. Effect of Penalty Weight

Table III summarises the penalty sweep for Condition 2 on the corridor map (3 seeds, 500 steps). With the IG-weighted scoring of (6), $w_p = 10$ achieves 93% coverage with 164 disconnections — a 66% reduction versus pure frontier at only 1.1 percentage points of coverage cost. Above $w_p = 25$ the penalty exceeds typical IG values and coverage collapses as robots revert to frozen behaviour.

TABLE III
CORRIDOR MAP — COVERAGE AND DISCONNECTIONS VS. PENALTY WEIGHT w_p

w_p	Coverage (%)	Disconnections
0 (pure frontier)	94.1	483
5	93.1	290
10	93.0	164
25	90.1	480
50	26.5	0

VII. DISCUSSION

Coverage–connectivity trade-off. The soft penalty at $w_p = 10$ identifies a practical Pareto point: near-full coverage with substantially fewer disconnections than unconstrained exploration. The physical interpretation is clear — a disconnection is only accepted when the target frontier contains more than $w_p = 10$ unknown cells, i.e., when exploration value outweighs connectivity cost.

Map topology as a hard constraint. The rooms map reveals that some coverage–connectivity trade-offs are geometrically unavoidable rather than algorithmically controllable. When rooms are separated by more than r_{comm}, full coverage requires robots to enter isolated regions. No finite penalty can prevent this without sacrificing coverage — a finding that motivates Condition 3's convoy strategy as the only true zero-disconnection option.

Formation overhead. Adaptive formation (Condition 4) introduces a modest exploration delay of ≈19–65 extra steps to 90% relative to Conditions 1 and 2, with one seed in three failing to reach 90% in the corridor map. This overhead reflects cohesion-action overrides that redirect robots toward the group centroid instead of the highest-IG frontier. The delay is smallest in open maps (151 vs. 112 steps, Condition 1), where robots are rarely far from the centroid. On the connectivity side, the cohesion override fires before the soft penalty scorer, so Condition 4 incurs more disconnections than Condition 2 (381 vs. 164 on corridor) even though both use the same $w_p = 10$ penalty.

Adversarial robustness. Condition 5 shows that one defecting robot reduces corridor coverage by ≈28 percentage points and rooms coverage by ≈52 points relative to the unconstrained baseline (Condition 1: 94.1%→66.6%, 98.8%→46.4%). The open map is more resilient (26-point reduction) because wider free space allows other robots to continue exploring despite the adversary.

Limitations. The Chebyshev-distance communication model ignores wall occlusion; two robots in adjacent rooms may be within r_{comm} but unable to communicate. The planner uses a greedy per-robot ordering in the soft penalty that does not globally optimise the action selection. Future work should explore joint optimisation, topology-aware communication models, and learned policies.

Future directions. Several extensions follow naturally from this work. First, the discrete soft penalty could be replaced by a Control Barrier Function (CBF) formulation in continuous space, enforcing $\lambda_2 > 0$ as a hard safety constraint via a QP solved at each timestep [8]. This would provide formal guarantees rather than a heuristic penalty. Second, the Chebyshev communication model ignores wall occlusion; a line-of-sight or signal-propagation model would make connectivity estimates more realistic for indoor environments. Third, the greedy per-robot action ordering could be replaced by joint optimisation (e.g., mixed-integer linear programming or a learned multi-agent policy) to find globally optimal team actions rather than locally optimal ones. Fourth, heterogeneous teams with dedicated relay robots — whose sole role is to maintain the communication chain while explorer robots spread freely — could resolve the geometric room-separation constraint identified in Section VI-D. Finally, validation on physical hardware with real WiFi radios would expose noise, actuator error, and latency effects absent from the discrete grid simulator.

VIII. CONCLUSIONS

We presented a framework for connectivity-constrained multi-robot exploration comparing five conditions — from unconstrained frontier exploration to adversarial formation control. Three algorithmic contributions (IG-weighted soft penalty, multi-pass connectivity filter, and deadlock detection with cluster-skip) raised coverage from $\leq 32\%$ under the original hard connectivity veto to $\geq 93\%$ on corridor and open maps while cutting disconnection events by up to 66%. Rigid formation remains the only condition that guarantees zero disconnections, at the cost of reduced coverage due to its convoy structure. Map topology is identified as a key factor: room separation beyond r_{comm} creates an irreducible geometric connectivity constraint that no behavioral algorithm alone can resolve. These results suggest that soft, information-theoretic connectivity penalties — rather than hard graph-theoretic vetos — offer a principled and practical approach to the coverage–connectivity trade-off in structured environments.

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